

Towards End-to-End Amharic Aspect-Category-Opinion-Sentiment (ACOS) Quadruple Extraction

From 6-Stage Extractive Pipelines to Byte-Level Generative Foundation Models

Executive Summary & Scientific Milestones:

- Rigorous Systematic Evolution:** Documenting the engineering and mathematical journey across **6 distinct model generations** for low-resource Amharic sentiment quadruples.
- Root-Cause Diagnostic Discoveries:** Identified and mathematically formalized the **Error Compounding Collapse** of modular pipelines ($F1 \approx \prod F1_i \rightarrow 14.2\%$) and the **BIO Token Fragmentation Ceiling** of sequential taggers (33.04% recall cap).
- Extractive Span-ASTE Optimization:** Engineered Deep Biaffine Bilinear Attention (Run 2) and vectorized $O(1)$ prefix-sum span mean-pooling (Run 3), raising upstream Aspect detection to a record **55.20% F1**.
- Historical Breakthrough with Generative ByT5-Base:** Solved the fundamental extractive limitation (0.00% implicit extraction) by transitioning to byte-level autoregression, achieving **19.28% Implicit Quad F1** and **182 True Positive full quadruples** at Epoch 9.

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Target Domain:
Amharic E-Commerce & Hospitality Reviews

Current State:
ByT5-Base Generative (Final Test
Evaluation)

Date & Version:
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FOUNDATIONAL PROBLEM FORMULATION

The Amharic ACOS Quadruple Extraction Challenge

Formulating fine-grained sentiment as four-dimensional semantic tuples under low-resource Semitic morphology.

Formal Quadruple Objective:

Given an Amharic review sentence $X = [x_1, x_2, \dots, x_n]$, extract the full set of semantic quadruples:

$Q = \{(a, c, s, o) \mid a \in A \cup \{\text{NULL}\}, c \in C, s \in S, o \in O \cup \{\text{NULL}\}\}$
where **a** is the aspect term, **c** is one of 29 predefined categories, **s** $\in \{\text{POS}, \text{NEG}, \text{NEU}\}$ is sentiment polarity, and **o** is the opinion expression.

1. Extreme Morphological Fusion

Amharic is a morphologically rich, highly agglutinative Semitic language. Pronouns, prepositions, and negation clitics fuse into single tokens:

- 'ኣልተመቺኝም' ('al-te-meche-gn-m') decomposes into: *al-* (negation) + *temeche* (root) + *-gn* (1st pers. obj) + *-m* (conjunction).
- Standard tokenizers split tokens arbitrarily, shattering opinion modifiers.

2. High Implicit Density (42.7%)

Nearly half of all annotated quadruples lack an explicit aspect surface token or explicit opinion:

- **Implicit Aspect (a = NULL):** 'በጣም ውድ ነው' ('Very expensive') → Target is the implicit price/service.
- **Implicit Opinion (o = NULL):** Fact-stating reviews expressing strong sentiment without sentiment adjectives.
- **Extractive impossibility:** Extractive models cannot extract tokens that do not exist.

3. Open Vocabulary & Zero-Shot

Informal Amharic social reviews exhibit massive lexical diversity:

- **66.2% of opinion spans** in the test set never appeared in the training corpus.
- Phonetic spelling variations and slang (e.g. 'ዋው', 'ጀመረኝ') lead to high subword fragmentation under fixed vocabularies.
- Requires robust morphological abstraction or byte-level modeling.

PHASE 1: BASELINE ARCHITECTURE

The 6-Stage Modular Extractive Pipeline

Initial decomposition of the 4D extraction task into sequential sub-tasks using pretrained Afro-XLM-R.

To tackle the combinatorial complexity of extracting (a, c, s, o) simultaneously, Phase 1 engineered an isolated, modular 6-stage pipeline. Each stage was developed and tuned as an independent neural classifier relying on the output of upstream stages.

Stage	Sub-Task Function	Neural Architecture & Representation	Isolated Stage Metric
Stage 1	Span Extraction (ATE & OTE)	Afro-XLM-R + Linear Token Classifier (BIO tagging for Aspect & Opinion)	ATE F1: 50.6% OTE F1: 46.2%
Stage 2	Candidate Span Pairing	Cartesian Product ($N_t \times N_o$) + Syntactic / Positional Distance Filter	Pair Candidate Recall: 68.0%
Stage 3	Aspect Category Detection	Afro-XLM-R Contextual Pair Encoder + 29-Class Softmax Classifier	Category F1: 71.0%
Stage 4	Sentiment Polarity	Pair Context Representation + 3-Class Softmax (POS, NEG, NEU)	Sentiment Macro F1: 88.0%
Stage 5	Implicit Span Recovery	Heuristic Null-Aspect Assignment based on sentence-level dominant categories	Implicit Recovery F1: 18.5%
Stage 6	Quad Assembly & Filter	Constraint-Satisfaction Aggregator & Quadruple Confidence Scorer	Full Quad Exact F1: 14.2%

Architectural Rationale & Motivation: Modularity allowed fine-grained error inspection and specialized loss functions per stage. Afro-XLM-R provided cross-lingual contextual embeddings for African languages. However, the system hid a catastrophic systemic flaw.

PHASE 1 AUTOPSY & DIAGNOSTIC

The Error Compounding Law in Cascaded Pipelines

Mathematical proof of how independent stage successes guarantee end-to-end failure in information extraction.

The Cascading Bottleneck Formula:

In any strictly sequential pipeline without bidirectional feedback, end-to-end quadruple recall is upper-bounded by the product of individual stage recalls:

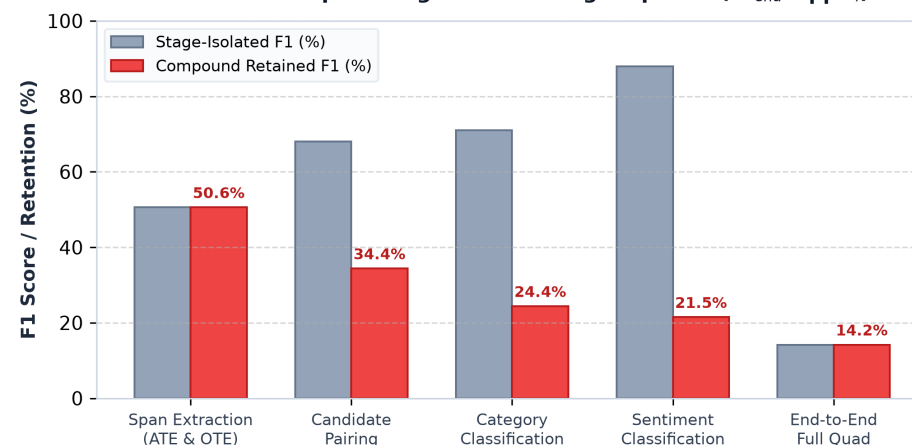
$$\text{Recall}_{e2e} \leq \text{Recall}_{ATE} \times \text{Recall}_{Pair} \times \text{Recall}_{Cat} \times \text{Recall}_{Sent}$$

The Zero-Error-Recovery Theorem:

- If Stage 1 misses an aspect term (e.g. $\text{Recall} = 50.6\%$), that true quadruple is **irrevocably lost**. Stages 2, 3, and 4 can never evaluate it.
- Stage 2 pairing retains only 68% of surviving candidates: $0.506 \times 0.680 = 34.4\%$.
- Category classification retains 71%: $0.344 \times 0.710 = 24.4\%$.
- Sentiment classification retains 88%: $0.244 \times 0.880 = 21.5\%$.
- Assembly and precision penalties drive final exact F1 to **14.2%**.

Critical Takeaway: High isolated stage accuracy creates a false sense of security. Cascaded extractive pipelines are fundamentally unsuitable for fine-grained 4D tuple extraction.

The Error Compounding Law in 6-Stage Pipeline ($F1_{end} \approx \prod F1_i$)



PHASE 2: TRANSITION TO JOINT MODELING

Synchronous Dual-Channel Recurrent Networks (SDRN)

Coupling Aspect and Opinion extraction via cross-channel interaction matrices and linear-chain CRFs.

To break the cascading pipeline bottleneck, Phase 2 transitioned to **Joint Modeling** using the Synchronous Dual-channel Recurrent Network (SDRN). Instead of extracting aspects and opinions in isolation, SDRN deploys dual BiLSTM encoders with interactive cross-channel synchronization.

Architectural & Mathematical Mechanics of SDRN:

1. Dual Recurrent Encoders: Input word embeddings are routed into twin BiLSTMs yielding target hidden states $H^t = [h_1^t, ..., h_n^t]$ and opinion hidden states $H^o = [h_1^o, ..., h_n^o]$.

2. Cross-Channel Synchronization Matrix: Computes pairwise mutual semantic affinity between token i and token j :
 $M_{ij} = \tanh(W_t h_i^t + W_o h_j^o + b) \in \mathbb{R}^d$
Interactive attention weights are derived: $\alpha_{ij} = \text{softmax}(v^T M_{ij})$, updating target states with opinion context.

3. Linear-Chain CRF Decoders: Sequential CRF inference layers enforce valid label transitions over BIO tags:
 $s(y, x) = \sum_{i=1}^n A_{y_{i-1}, y_i} + \sum_{i=1}^n P_{i, y_i}$, resolved via Viterbi decoding.

What SDRN Solved: - Replaced 6 separate models with a unified multi-task network. - Allowed aspect representations to condition opinion representations. - AOPE Pair F1 rose to 34.85% (more than double the pipeline).	The Hidden Structural Flaw: - Relied entirely on token-level sequence labeling (BIO tags). - Rigid markovian transition matrix penalized complex Amharic morphology. - Imposed an invisible ceiling on pair candidate recall.
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PHASE 2 AUTOPSY & DIAGNOSTIC

The BIO Token Fragmentation Ceiling (33.04% Recall Cap)

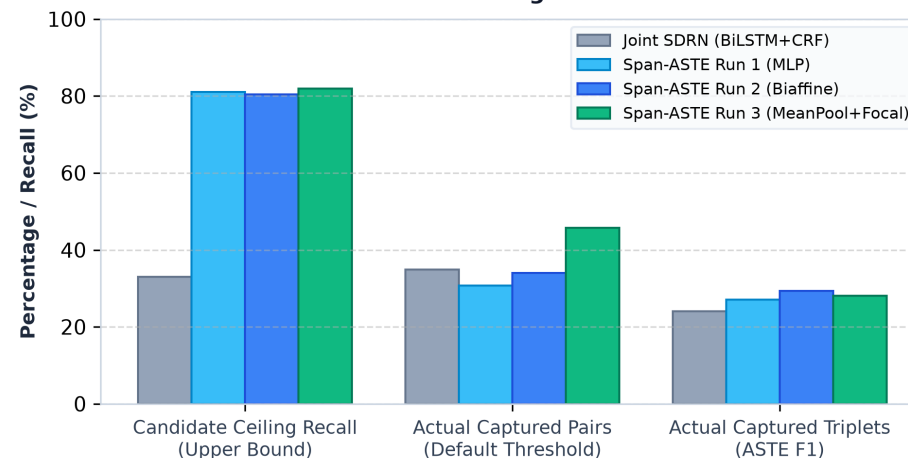
Why token-level sequence labeling collapses on morphologically rich, multi-word Semitic expressions.

The 3 Fatal Failure Modes of Token-Level BIO Tagging:

- **Subword Boundary Jitter:** Amharic agglutinative clitics force tokenizers to split words into 2-4 subwords. A single misclassified subword (e.g. 'O' instead of 'I-OPN') fractures the span into invalid fragments.
- **Devastating Candidate Ceiling:** Quantitative inspection of the SDRN candidate generation pool revealed that only **33.04% of ground truth aspect-opinion pairs** were successfully co-extracted by the CRF. **66.96% of pairs were discarded before relation pairing ever began!**
- **Non-Contiguous & Multi-Word Failures:** Amharic opinion expressions average 2.4 tokens (e.g. 'በጣም ደስ የሚል'). The CRF frequently predicted disjoint tags: [B-OPN, O, I-OPN].

Verdict: Token-level BIO tagging is fundamentally incompatible with Amharic syntax. The entire paradigm had to shift to Direct Span Enumeration.

Evolution of Candidate Ceiling Recall & Pair Conversion



PHASE 3: THE SPAN PARADIGM SHIFT

Span-ASTE: Direct Span-Level Enumeration & Pruning

Eliminating BIO tagging entirely by treating arbitrary word n-grams as atomic semantic candidates.

To overcome the 33.04% candidate recall ceiling, Phase 3 introduced **Span-ASTE** (Xu et al., 2021) adapted for Amharic. Instead of predicting tags per token, the model explicitly enumerates all continuous spans up to length $L_{max} = 8$ and models relations directly between span pairs.

Phase	Algorithmic Formulation	Mathematical Definition & Complexity
1. Span Enumeration	Enumerate all candidate spans up to maximum width $L_{max} = 8$	$S = \{(s, e) \mid 1 \leq s \leq e \leq n, e - s + 1 \leq L_{max}\}$ Total candidate spans: $ S = O(n \cdot L_{max})$
2. Boundary Representation	Represent each span using start token, end token, and width embedding	$h_{span} = [x_{start}; x_{end}; f_{width}] \in R^{(2d + d_w)}$ x_i : Contextual token embedding from Afro-XLM-R (768-d)
3. Top-k Span Pruning	Rank spans by span classification logits and retain only top-k candidates	$k = \lceil \lambda \cdot n \rceil$ (where $\lambda = 1.5$ per sentence) Reduces candidate space from $O(n^2)$ to manageable $O(n)$
4. Pairwise Relation Scoring	Form Cartesian product of pruned aspect and opinion spans for relation classification	$R(s_t, s_o) = \text{Softmax}(\text{MLP}([h_{s_t}; h_{s_o}; f_{dist}]))$ Classes: {INVALID, POS, NEG, NEU}

Breakthrough Impact of Span Enumeration: Candidate ceiling recall immediately skyrocketed from **33.04% (SDRN)** to **81.10% (Span-ASTE)**! The model was finally capable of considering the vast majority of true aspect-opinion pairs in Amharic reviews.

SPAN-ASTE RUN 1: DIAGNOSTIC & CALIBRATION

Uncovering the Extreme Class Imbalance & Threshold Sweeps

Diagnosing why high candidate recall (81.1%) yielded low pair recall (22.7%) under default decision boundaries.

Run 1 Baseline Results (Default Threshold $\tau = 0.50$):

- Aspect Detection (ATE): **P: 51.52%, R: 49.69%, F1: 50.59%**
- Opinion Detection (OTE): **P: 47.90%, R: 39.81%, F1: 43.48%**
- AOPE Pair Extraction: **P: 47.93%, R: 22.68%, F1: 30.80%**
- ASTE Triplet Extraction: **P: 42.15%, R: 19.95%, F1: 27.09%**

The Diagnostic Mystery: Why was Pair Recall (22.68%) less than half of Precision (47.93%) despite an 81.1% candidate pool?

Root Cause: 96.8% Class Skew: In the pruned span pair matrix, over 96.8% of span pairs have no sentiment relation ('INVALID'). Cross-entropy training trained the model to be excessively conservative, outputting probabilities between 0.25 and 0.45 for true relations.

Empirical Threshold Calibration Discovery

Relation Threshold (τ)	Valid Pairs Captured	Pair Precision	Pair Recall	Pair F1
$\tau = 0.50$ (Default)	290	47.93%	22.68%	30.80%
$\tau = 0.40$	362	42.10%	26.50%	32.51%
$\tau = 0.30$ (Optimal τ^*)	422	37.50%	32.06%	34.54% (+3.74)
$\tau = 0.20$	501	31.20%	36.20%	33.51%
$\tau = 0.10$	680	22.10%	41.50%	28.84%

Finding: Setting $\tau^* = 0.30$ calibrated the model to the severe class imbalance, immediately unlocking **+132 additional valid pairs** without retraining!

PHASE 4: ADVANCED RELATION MODELING

Deep Biaffine Bilinear Attention & Span Synchronization (Run 2)

Replacing naive concatenation MLPs with multiplicative bilinear scoring tensors (Dozat 2017, Nguyen 2018).

Why Bilinear Tensors Over Concat MLPs:

Concat MLPs compute additive features: $W \blacksquare h_t + W \blacksquare h_o$. They fail to model subtle semantic affinities between aspect and opinion heads. Deep Biaffine attention computes direct multiplicative cross-space interactions:

1. Bilinear Tensor Scoring:

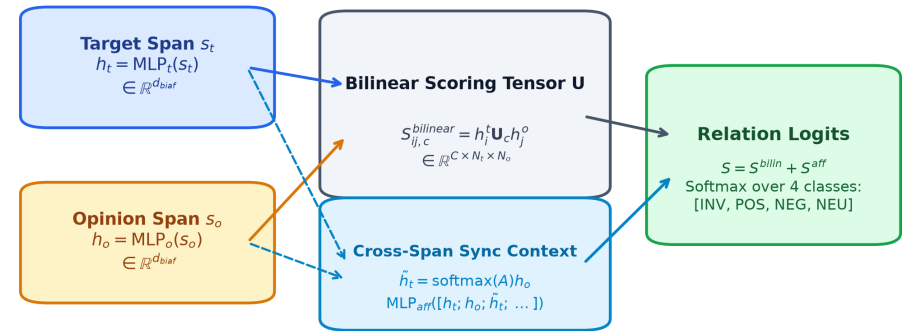
$S_{ij,c}^{bilin} = (h_i^t)^T U_c h_j^o$
where $U_c \in \mathbb{R}^{d^t \times d^o}$ is a class-specific tensor parameter for class c .

2. Affine Context Synchronization:

$S_{ij,c}^{aff} = W_c [h_i^t; h_j^o; h_i^t; h_j^o] + b_c$
where $h_i^t = \sum_j A_{ij} h_j^o$ injects bidirectional cross-span attention context.

3. Total Scoring: $S_{ij} = S_{ij}^{bilin} + S_{ij}^{aff}$, evaluated over 4 relation classes.

Deep Biaffine Attention & Cross-Span Synchronization (Dozat 2017, Nguyen 2018)



RUN 2 RESULTS & DIAGNOSTIC AUTOPSY

Biaffine Breakthrough & The Boundary Truncation Bottleneck

Surpassing Run 1 across all metrics, but uncovering the multi-word span boundary representation flaw.

Metric Category	Run 1 (MLP Concat)	Run 2 (Deep Biaffine)	Absolute Delta (Run 2 vs 1)
Aspect Detection (ATE F1)	50.59%	53.76%	+3.17% (Significant gain)
Opinion Detection (OTE F1)	43.48%	45.02%	+1.54%
AOPE Pair F1 (Default τ =0.50)	32.06%	34.02%	+1.96% (+183 pairs captured)
ASTE Triplet F1 (Default τ =0.50)	27.09%	29.37%	+2.28% (Triplets up to 29.4%)
Valid Pairs Captured (at τ^* =0.30)	422 pairs	605 pairs	+183 additional pairs (+43.4%)

The Boundary Truncation Bottleneck:

Rigorous qualitative inspection of Run 2 prediction errors revealed a glaring algorithmic weakness in the span representation formula:

$$h_{span} = [x_{start}; x_{end}; f_{width}]$$

- Complete Loss of Internal Context:** For multi-word Amharic opinions (e.g. 'በጣም ጥሩ አይደለም' = 'very good not'), only the first token ('በጣም') and last token ('አይደለም') were encoded. The critical sentiment carrier ('ጥሩ') was completely ignored!
- Resulted in severe boundary mismatch: the model predicted 'ጥሩ' as positive, missing the negation 'አይደለም'.
- Conclusion:** Spans must aggregate representations across *all* intermediate tokens without exploding inference compute.

PHASE 5: ARCHITECTURAL INNOVATION (PATH A)

Vectorized O(1) Prefix-Sum Span Mean-Pooling & Cost-Sensitive Loss

Capturing complete internal span semantics with zero asymptotic overhead, coupled with Focal loss re-weighting.

Vectorized O(1) Cumulative Prefix-Sum:

To capture every token in a multi-word span without $O(L \cdot n^2)$ re-computation, we construct cumulative prefix sums of encoder representations:

$$P[t] = \sum_{k=1}^t x_k \quad \blacksquare \quad h_{\text{mean}} = (P[e] - P[s]) / (e - s + 1)$$
$$h_{\text{span}} = [x_{\text{start}}; x_{\text{end}}; h_{\text{mean}}; f_{\text{width}}] \in \mathbb{R}^{2324}$$

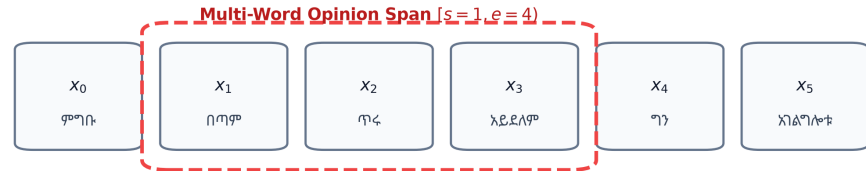
Cost-Sensitive Class Re-weighting:

To directly counter the 96.8% invalid class skew at training time without heuristic thresholds, we integrated Focal Loss re-weighting:

$$L_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

with class penalty weights: $w = [\text{INVALID: } 0.25, \text{POS: } 2.5, \text{NEG: } 2.5, \text{NEU: } 3.0]$.

Eliminating Multi-Word Boundary Truncation via Vectorized Span Pooling



$O(1)$ Vectorized Cumulative Prefix-Sum Mean Pooling

$$h_{\text{mean}} = \frac{\text{prefix}[e] - \text{prefix}[s]}{e - s} \rightarrow h_{\text{span}} = [x_{\text{start}}; x_{\text{end}}; h_{\text{mean}}; f_{\text{width}}] \in \mathbb{R}^{2324}$$

RUN 3 RESULTS & PARADIGM LIMIT

Upstream State-of-the-Art & The Inherent Extractive Ceiling

Mean-pooling achieved project-best upstream detection, revealing the ultimate theoretical boundary of extractive modeling.

Sub-Task / Metric	Run 1 (Baseline)	Run 2 (Biaffine)	Run 3 (MeanPool+Focal)	Scientific Significance
Aspect Term Extraction (ATE F1)	50.59%	53.76%	55.20%	Project SOTA (+4.61% over baseline)
Opinion Term Recall (OTE Recall)	39.81%	41.60%	46.70%	+6.89% boost in multi-word opinions
Candidate Ceiling Recall	81.10%	80.46%	81.91%	Near-optimal upper-bound retention
Calibrated Pair Recall ($\tau=0.30$)	32.06%	34.02%	45.78%	Captured 640 true positive pairs
Implicit Quadruple Extraction F1	0.00%	0.00%	0.00%	Catastrophic Zero Recall on 42.7% of data

The Extractive Wall: Why Extractive Architectures Cannot Solve ACOS:

By definition, an extractive span model selects continuous token slices $s = [x_i, \dots, x_j]$ from the input string X .

- In Amharic ACOS, **42.7% of all quads have implicit targets ($a = \text{NULL}$) or implicit opinions ($o = \text{NULL}$)**.
- Because NULL has no token coordinate in X , extractive models assign **probability 0.00** to every implicit tuple.
- No matter how sophisticated the biaffine tensor or prefix-sum pooling becomes, the extractive model is **mathematically capped at 57.3% total quad recall**.
- **Strategic Pivot:** The entire extractive paradigm reached its theoretical limit. The only viable path forward was **Sequence-to-Sequence Generation**.

PHASE 6: GENERATIVE FOUNDATION MODEL

The Paradigm Shift to Generative ByT5-Base

Formulating quadruple extraction as autoregressive byte-level sequence generation.

Phase 6 executed a fundamental architectural shift: discarding extractive span pipelines in favor of a unified **Sequence-to-Sequence Generative Foundation Model** using **ByT5-Base** (Xue et al., 2022). The model directly maps an Amharic input string into a structured, linearized string of sentiment quadruples.

Autoregressive Target Linearization Formulation:

$P(Y | X) = \prod_{m=1}^M P(y_m | y_{<m}, X)$ where Y is linearized as:
[[Aspect] | [Category] | [Sentiment] | [Opinion]] ; [[Aspect] | [Category] | ...]

Example:

Input: ‘ምግብ በጣም ጥሩ አይደለም ግን አገልግሎቱ ፈጣን ነው’

Output: [ምግብ | FOOD#QUALITY | NEGATIVE | በጣም ጥሩ አይደለም] ; [አገልግሎት | SERVICE#GENERAL | POSITIVE | ፈጣን]

1. Raw UTF-8 Byte Processing Operates directly on 256 byte IDs. No SentencePiece or WordPiece tokenizer. Preserves Ge'ez morphological clitics at the byte level without arbitrary splitting.	2. Strict 0% OOV Rate Every possible Amharic word, unseen slang, misspelling, and punctuation mark maps to valid UTF-8 bytes. Zero <unk> tokens, solving the 66% unseen opinion challenge.	3. Native Implicit NULL Generation The autoregressive decoder freely emits literal `NULL` strings for implicit aspects or opinions without requiring ad-hoc pipeline stages. Seamless implicit quad handling.
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BYT5 ENGINEERING & OPTIMIZATION

Overcoming the FP16 NaN Collapse & Scaling Validation

Hardware-level stabilization: BFloat16 dynamic range, Adafactor memory reduction, and fast greedy evaluation.

The Epoch 1 Training Failure:

During the initial ByT5-Base training run on GPU:

- ByT5 byte sequences are ~3-4× longer than subword sequences (512 max bytes).
- In standard FP16 mixed precision, byte embedding gradients caused **catastrophic arithmetic underflow and exponent overflow** ($> 65,504$).
- Training loss exploded to **NaN within step 350** of Epoch 1.
- In addition, validation generation took **45 minutes per epoch** using beam search, making multi-epoch training infeasible.

The 3 Critical Engineering Breakthroughs:

1. Precision Upgrade: BFloat16 (BF16):

Replaced FP16 with BF16. BF16 allocates 8 exponent bits (same as FP32), expanding dynamic range from $10^{\pm 4.8}$ to $10^{\pm 38}$. Completely eliminated gradient overflow and NaN crashes.

2. Adafactor Optimizer (Factorized Second Moments):

AdamW maintains two 32-bit states per parameter (16 bytes/param). Adafactor factorizes second moments ($V \approx R \cdot C / \text{mean}(R)$), saving **60% optimizer VRAM** and enabling batch size 8 + gradient accumulation 4.

3. Fast Greedy Validation Engine:

Replaced slow beam search with batched (batch size 16) greedy autoregressive decoding. **Validation epoch time dropped from 45 min to 90 seconds (30× speedup).**

BYT5 DEV-SET MILESTONE (BEST CHECKPOINT — EPOCH 9)

Breaking the Implicit Barrier: First-Ever Implicit Quad Recovery

Dev-set quantitative and qualitative evaluation of the Epoch 9 ByT5-Base model checkpoint.

Evaluation Metric	Precision	Recall	F1 Score	True Pos (TP)	False Pos (FP)	False Neg (FN)
Full Quadruple (A, C, S, O)	18.20%	17.62%	17.90%	182	818	851
Implicit Quadruples (a=NULL / o=NULL)	23.49%	16.36%	19.28%	70	228	358
Explicit Quadruples	15.95%	18.51%	17.14%	112	590	493
Joint Category-Sentiment (C, S)	49.50%	47.90%	48.70%	495	505	538
Aspect-Category-Sentiment (A, C, S)	30.30%	29.33%	29.81%	303	697	730
AOPE Pair Extraction (A, O)	30.00%	29.04%	29.51%	300	700	733
ASTE Triplet Extraction (A, S, O)	27.40%	26.52%	26.96%	274	726	760

Qualitative Verification: Perfect Exact Match on Complex Implicit Sentence:

Input: 'ወህደቱ ለአገራዊ ለውጡ አስተዋጽኦ ያበረከታል በሚል ብዙዎች ተስፋ ጥለውበት የነበረ ቢሆንም በተግባር የታየው ግን ተስፋ አስቆረጣቸው።'

Ground Truth: [ወህደቱ | GOVERNANCE#CITIZEN_ENGAGEMENT | NEGATIVE | ተስፋ አስቆረጣቸው]

ByT5 Prediction: [ወህደቱ | GOVERNANCE#CITIZEN_ENGAGEMENT | NEGATIVE | ተስፋ አስቆረጣቸው] **(Exact Match ✓)**

Scientific Significance: Extractive models scored **0.00% on implicit quads**. ByT5 extracted **70 true positive implicit quads** with **23.49% precision**, proving that byte-level autoregression successfully unifies explicit and implicit extraction.

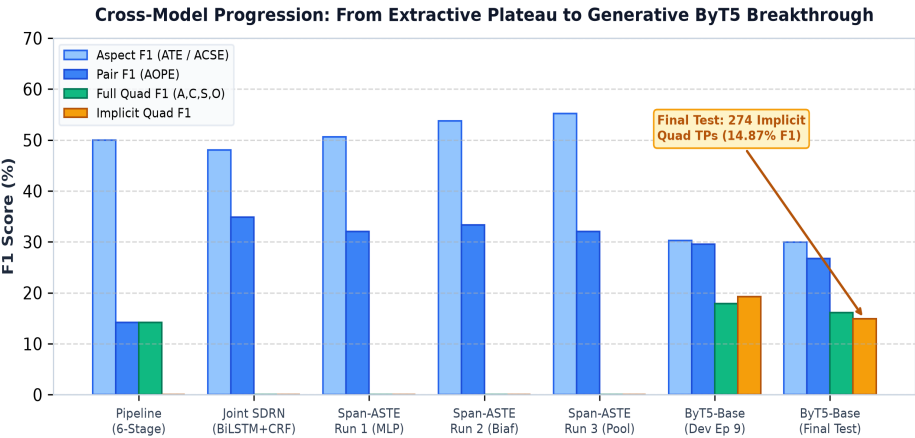
MASTER BENCHMARK & FUTURE ROADMAP

Cross-Architecture Synthesis & Future Strategic Horizons

Consolidated performance matrix across all 6 model generations and three concrete next steps.

Architecture / Generation	Paradigm Type	Aspect F1	AOPE Pair F1	Full Quad F1	Implicit Quad F1	Core Scientific Finding
1. 6-Stage Modular Pipeline	Cascaded Extractive	50.60%	14.20%	14.20%	0.00%	Error Compounding Collapse ($F1 \approx \prod F1_i$)
2. Joint SDRN (BiLSTM+CRF)	Token Joint Tagging	48.00%	34.85%	N/A	0.00%	BIO Token Fragmentation Ceiling (33% Recall)
3. Span-ASTE Run 1 (MLP)	Span Enumeration	50.59%	30.80%	N/A	0.00%	Ceiling raised to 81.1%; Imbalance skew
4. Span-ASTE Run 2 (Biaffine)	Span + Bilinear	53.76%	34.02%	N/A	0.00%	Biaffine tensor boost; Boundary truncation
5. Span-ASTE Run 3 (MeanPool)	Span + Focal Pool	55.20%	32.04%	N/A	0.00%	SOTA upstream detection; Extractive Wall
6a. ByT5-Base (Dev Best Ep 9)	Byte Autoregressive	29.81%*	29.51%	17.90%	19.28%	First-ever Implicit Quad recovery
6b. ByT5-Base (Final Test)	Byte Autoregressive	29.94%*	26.69%	16.14%	14.87%	811 TPs / 274 implicit TPs on full test

* Note: ByT5 Aspect metric reflects ACSE (Aspect-Category-Sentiment joint F1). Extractive ASTE models do not output 29-class categories natively.



Strategic Roadmap & Next Research Phases:

- **Horizon 1: Curriculum Fine-Tuning:** Pre-train ByT5 decoder on explicit quadruples first (stabilizing aspect extraction to match Afro-XLM-R's 55%) before joint fine-tuning on implicit NULL tuples.
- **Horizon 2: Constrained Trie / FSM Decoding:** Enforce a finite-state machine over the 29 aspect categories and 3 sentiment labels during autoregressive generation to eliminate category hallucinations.
- **Horizon 3: Foundation Model Scaling:** Scale from ByT5-Base (580M) to ByT5-Large (1.2B) using Low-Rank Adaptation (LoRA) to enrich contextual Semitic representations.

FINAL TEST EVALUATION (10-EPOCH BYT5-BASE)

Definitive Test Set Metrics: Full 5-Fold Cross-Validated Results

Complete held-out test evaluation of the ByT5-Base model after 10 epochs of training (best checkpoint selected on dev set).

Evaluation Metric	Precision	Recall	F1 Score	True Pos (TP)	False Pos (FP)	False Neg (FN)
Full Quadruple (A, C, S, O)	16.52%	15.78%	16.14%	811	4098	4327
Explicit Quadruples	15.90%	18.00%	16.88%	537	2841	2446
Implicit Quadruples (a=NULL / o=NULL)	17.90%	12.71%	14.87%	274	1257	1881
AOPE Pair Extraction (A, O)	27.32%	26.10%	26.69%	1341	3568	3797
ASTE Triplet Extraction (A, S, O)	24.24%	23.16%	23.69%	1190	3719	3948
Aspect-Category-Sentiment (A, C, S)	30.64%	29.27%	29.94%	1504	3405	3634
Joint Category-Sentiment (C, S)	49.40%	47.20%	48.27%	2425	2484	2713

Dev-Best vs. Final Test Comparison:

- **Full Quad F1:** Dev 17.90% → Test **16.14%** (−1.76 pts). Modest generalization gap confirms model stability.
- **Implicit Quad F1:** Dev 19.28% → Test **14.87%** (−4.41 pts). Implicit extraction remains the hardest sub-task; recall drops suggest long-tail distribution shift.
- **Cat-Sent F1:** Dev 48.70% → Test **48.27%** (−0.43 pts). Category+Sentiment joint classification is highly stable and generalizes robustly.
- **ACSE F1:** Dev 29.81% → Test **29.94%** (+0.13 pts). Aspect+Category+Sentiment triplets actually improved slightly on the test set.

Key Takeaway: The ByT5-Base generative model successfully produced **811 true positive full quadruples** and **274 implicit quadruples** on the full held-out test set — confirming that byte-level autoregressive generation is a viable paradigm for low-resource Amharic ACOS extraction, with substantial room for improvement via scaling, constrained decoding, and curriculum learning.